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Abstract

In 2017, the GRADE (Grading of Recommendations Assessment, Development and Evaluation) working group defined the certainty of evidence as the certainty that the true effect lies on one side of a threshold or in a particular range. This definition has proved useful as the basis for rating certainty, facilitating the interpretation of the results for the target audience. However, the categorization of suggested thresholds and ranges as levels of contextualization led to inconsistencies between the initial and subsequent papers and has proved confusing for some GRADE users.

Although considering context in choosing thresholds remains worthwhile, the GRADE working group will no longer use the categorization of contextualization. It will instead refer simply to chosen thresholds or ranges for determining the target of certainty rating.

In 2017, the GRADE (Grading of Recommendations Assessment, Development and Evaluation) working group published a key paper that changed GRADE's appr rating the certainty of evidence (1). We had previously advised that authors rat certainty in the effect estimate. The 2017 guidance advised GRADE users to rate their certainty that the true effect lies on one side of a threshold or in a particular range. It has subsequently become clear that specifying the target of the certainty rating (that

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is, what one is rating one's certainty in) in relation to thresholds is an essential first step before rating the certainty of evidence for individual outcomes. The Figure shows commonly used thresholds to support ratings of certainty for benefit and harm of individual outcomes.

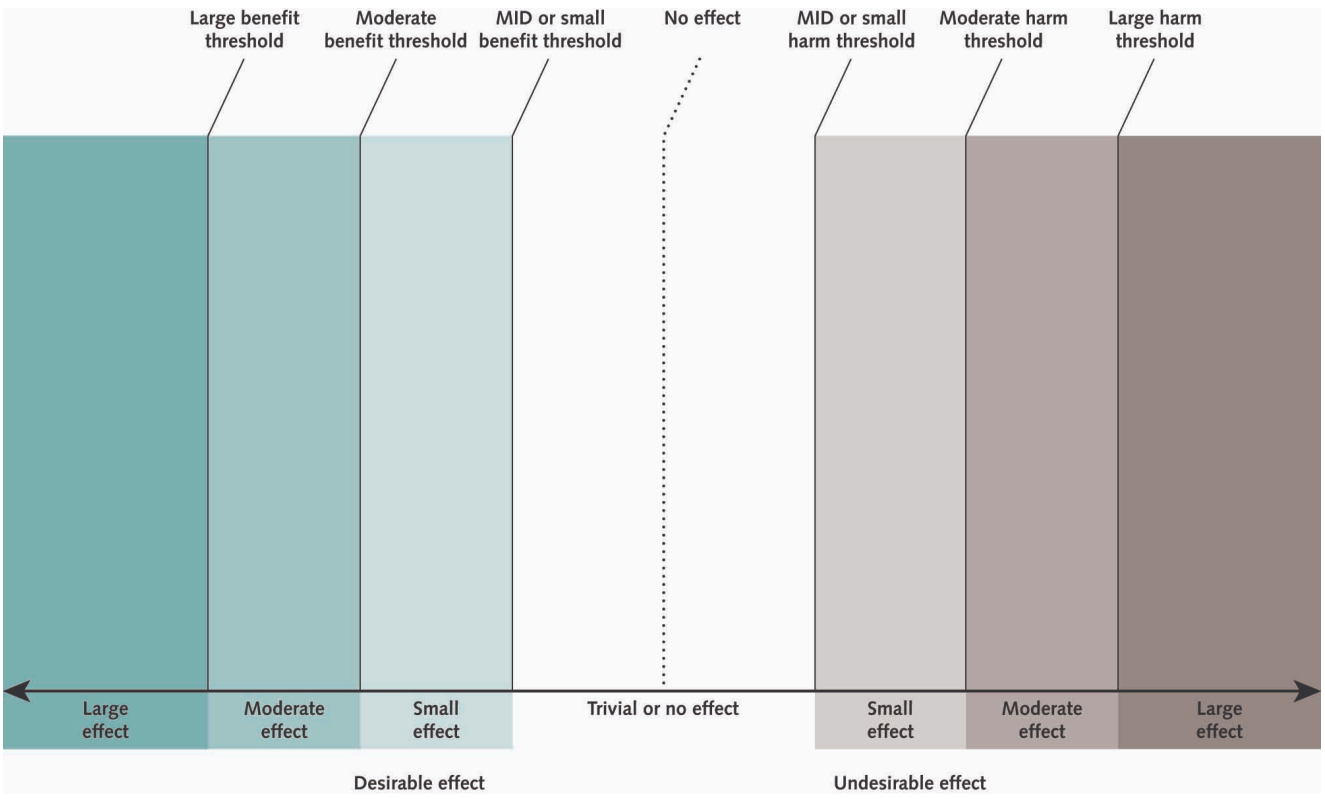

Figure. Thresholds and ranges that one might use when rating certainty of evidence for an individual outcome.

MID = minimal important difference.

The 2017 clarification paper suggested possible thresholds and used the term "contextualization" to label their categorization. "Noncontextualized" referred t choice of the null as the threshold, with the note that the choice of this thresh obviated the need for associated value judgments. "Partially contextualized" referred to thresholds considering the magnitude of effect (minimal important difference

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[MID], moderate and large effect thresholds) and "fully contextualized" to thresholds considering all people-important outcomes and consequences (net benefit).

Subsequent GRADE writings kept the notion of categories and continued using the term "contextualization." However, the categorization and associated terminology changed. We noted that even when users choose the null as the threshold, they have to make a value judgment about the choice of the outcomes on which to focus. We therefore modified the terminology and classification, referring to "minimally contextualized" when authors have chosen the null or MID and "partially contextualized" when they have chosen to use ranges. These terms appeared in the titles of GRADE working group papers (2–5).

Because they were perceived as nonintuitive, these categories of contextualization and change in terminology have confused some GRADE users. Contextualization also has other connotations in guidelines (for example, when they are implemented across settings [6] and when referring to "contextual factors," such as acceptability and feasibility [7]). The contextualization categories may also draw focus from the key issue of defining the target of the certainty rating and specifying the thresholds used. The Appendix (available at Annals.org) provides specific examples of the limitations of the categorization.

The objective of the current article is therefore to offer guidance that resolves the confusion related to the categories of contextualization for GRADE users involved in a systematic review, health technology assessment, coverage decision, or health guideline and for educators teaching GRADE to clinical audiences.

Methods

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This guidance was developed though iterative discussions within the GRADE of evidence project group (with more than 100 members) and the entire GRADE working group, according to the methods for developing official GRADE working group papers (8). A smaller core group drafted the paper with discussions and

feedback from the wider project group on several occasions. A draft version of the paper was presented and discussed with all attendees at the GRADE working group meeting in May 2024 and was presented for final approval in September 2024 before approval by the GRADE guidance group.

Results

We have concluded that it is advisable, when rating the certainty of evidence, to drop the "contextualization" categorizations altogether and rather refer only to the thresholds and the associated targets of certainty rating. This is an important change in the use of terminology, but the essential concept—the process of choosing thresholds and then using point estimates to determine the target—remains unchanged. Thus, it remains worthwhile to consider the extent to which context is taken into consideration in choosing thresholds, but the categorization into degrees of contextualization will no longer appear in GRADE working group papers (Table).

Table. Thresholds and Targets of Certainty Ratings and the Now-Abandoned Terminology for the Categories of These Thresholds

Consideration	Previous, Now-Abandoned Terminology	Thresholds to Be Used and Possible Targets of Certainty Ratings
Evidence synthesis authors want to avoid judgments around importance of the outcome	Initially "noncontextualized" and subsequently "minimally contextualized"	One threshold of "no effect" with targets of nonzero benefit and nonzero harm
GRADE users interested in whether there is trivial or no (i.e., unimportant) effect or an important effect	Initially "partially contextualized" and subsequently "minimally contextualized"	Thresholds of MID for benefit and MID for harm with 3 poss targets: trivial or effect, and impor benefit or harm
Decision makers (e.g., in guideline development or health technology assessment) or other	"Partially contextualized"	6 thresholds (MID or small effect, moderate, and large effect thresholds) with 7

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Consideration	Previous, Now-Abandoned Terminology	Thresholds to Be Used and Possible Targets of Certainty Ratings
GRADE users interested in whether there is trivial or no, small, moderate, or large effect		possible targets (trivial or no effect, and small, moderate, and large benefit or harm)

GRADE = Grading of Recommendations Assessment, Development and Evaluation; MID = minimal important difference.

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When choosing thresholds for rating certainty in an individual outcome, GRADE users can consider their remit and target audience (Wiercioch W, Morgan G, Piggott T, et al. GRADE guidance: using thresholds for judgments on health benefits and harms in decision-making. In preparation). For many health technology assessment contexts, clinical practice guidelines, and systematic reviews that are linked to a guideline, regardless of the certainty of the supporting evidence, authors must make judgments about the values of those whose care will be influenced by the recommendations. Trading off desirable and undesirable health effects associated with an intervention requires judging the relative importance of the outcomes.

The no-effect threshold can be useful when GRADE users want to avoid value judgments. Aside from the value judgment authors make when they choose the outcomes on which they will focus, the initial choice of no effect as the threshold involves no other value judgments. Systematic review authors may see it as beyond their remit to search for and interpret studies of values and preferences, or to collect new information on values. However, in cases where the point estimate is very near the no-effect threshold, authors may be compelled to make value judgments, their target from a benefit or harm to trivial or no effect (9).

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Systematic review authors may choose the MID or small effect threshold when they are writing for a clinical audience interested in whether an intervention yields an effect that people are likely to experience as important for an individual outcome. The MID or

small effect threshold represents the smallest difference in an individual outcome that people judge as important (either people in general, or those who are the relevant population of a systematic review, guideline, or health technology assessment). The MID or small effect threshold therefore clearly represents a value judgment.

Ranges of no or trivial, small, moderate, and large effects facilitate decision making using the evidence-to-decision frameworks from the GRADE working group (10). Rating the magnitude of effect of the individual outcomes can help inform the assessment of the net benefit or harm across all outcomes (4).

Discussion

This new GRADE guidance is more straightforward and facilitates accurate communication of the certainty ratings without changing the underlying concepts or methods. Although previous GRADE guidance papers use the now-rejected categorization, their discussion is completely consistent with the current focus on thresholds and targets (4, 5).

The approach to setting thresholds described in this article addresses individual outcomes. Nevertheless, guideline panels must ultimately consider whether the desirable consequences of an intervention (benefits) outweigh the undesirable consequences (harms and burdens) (4). Although guideline panels make judgments about their certainty in net benefit or net harm when they make strong or conditional recommendations, more explicit guidance on net benefit may prove useful (11).

Conclusion

The concept of contextualization remains useful, and consideration of the exte which context is taken into consideration in choosing thresholds remains wor however, a categorization of thresholds according to contextualization has proved more confusing than helpful. There are compelling reasons to consider each of the GRADE working group's suggested thresholds or ranges, depending on the

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circumstances. The GRADE working group will therefore no longer use the categorization according to contextualization but will instead refer simply to chosen thresholds or ranges for determining the target of certainty rating and assessing evidence certainty. Although this is an important change in terminology, the essential concept—the process of choosing thresholds and then using point estimates to determine the target—remains unchanged.

Appendix: Limitations to the Categories of Contextualization

Our experiences of the limitations of the categories of contextualization include the following.

First, GRADE users familiar with the original 2017 paper (1) but less familiar with subsequent use of the categories have repeatedly used the initial categorization or expressed confusion when confronted with the subsequent alternative.

Second, GRADE users have requested direction to a clear rationale and explanation for the change from the 2017 categories to the subsequent categories. Such an explanation does not exist, perhaps in part because differentiating a single value-based threshold from 3 value-based thresholds as the basis for categorization is very difficult to justify.

Third, in teaching prospective GRADE users about setting thresholds and targets of certainty rating, the categories have consistently proved unnecessary. Moreover, when we tried to incorporate the categorizations despite their being gratuitous, it impeded rather than facilitated the teaching.

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